



Software Testing and Quality Assurance
Second Semester 2025/2026
Homework Assignment # 1
Total Mark (20 points)

Task (A) – JUnit 5 Test Class

1. Create a JUnit 5 test class(s) that thoroughly tests the main behaviors of ProductStock, including:
 - Valid construction and invalid constructor parameters
 - Adding stock (addStock) – normal, boundary, and error cases
 - Reserving and releasing stock (reserve, releaseReservation)
 - Shipping reserved stock (shipReserved)
 - Removing damaged stock (removeDamaged)
 - Reorder logic (isReorderNeeded)
 - Updating thresholds and capacity (updateReorderThreshold, updateMaxCapacity)
2. Your test class must use *most* of the following JUnit 5 features:
 - Lifecycle: @BeforeAll, @BeforeEach, @AfterEach, @AfterAll
 - Basic test annotation: @Test
 - Timeout: @Timeout
 - Disable test: @Disabled (for a “future feature” test)
 - Tags: @Tag("sanity"), @Tag("regression")
 - Readability: @DisplayName
 - Optional: @Nested for grouping tests by scenario (e.g., “when stock is initialized”)
 - Optional: @ParameterizedTest with @ValueSource (e.g., for different addStock amounts)
3. Use appropriate JUnit assertion methods such as:
 - assertEquals, assertTrue, assertFalse
 - assertThrows for invalid inputs and illegal states
 - assertAll (optional) to group multiple related assertions

Task (B) – Test Case Design Sheet

Design a test case table for **ProductStock** including:

- Test Case ID
- Method / Feature under test
- Precondition(s)
- Input(s)
- Expected Result
- Type (e.g., normal, boundary, error)

You do not have to write code here – just the test design.

Task (C) – Coverage & Discussion

- Run your test suite with a coverage tool (e.g., IntelliJ, Eclipse, JaCoCo, etc.).
- Aim for at least 80% line coverage of **ProductStock**.
- In the discussion/oral part, be prepared to:
 - Show or describe the coverage report.
 - Explain which parts of the class are still not covered and why.
 - Justify how your test cases cover both happy paths and error paths (exceptions).